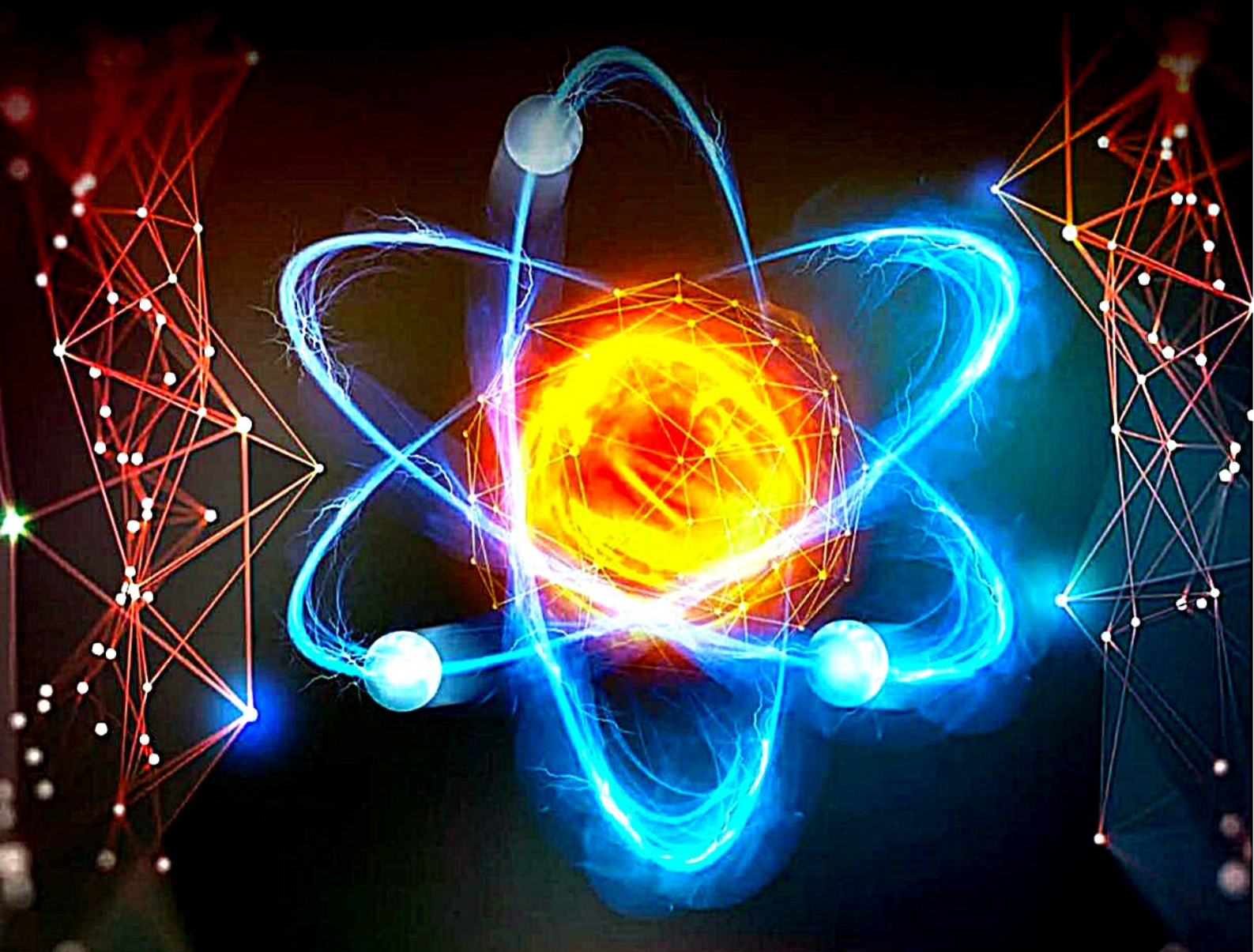


CLASS 12

PHYSICS

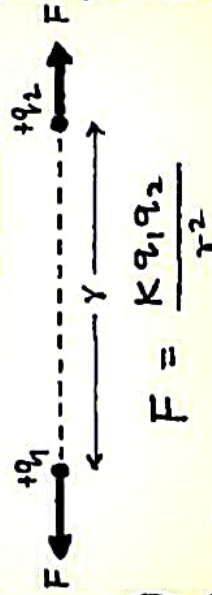
MIND MAPS



Basic Properties of Charge-

- 1.] Charge is quantised $Q = \pm ne$
 $n = \text{any integer value}$ $e = 1.6 \times 10^{-19} \text{ C}$
- 2.] Charge is conserved.
- 3.] Similar charges repel and dissimilar charges attract.

Coulomb's Law



$$F = \frac{k q_1 q_2}{r^2}$$

Electric field intensity

$$\vec{E} = \frac{kq}{r^2} \hat{r} \quad \vec{E} = \frac{\vec{F}}{q_0}$$

SI Unit - N/C , V/m

Dipole in a uniform Electric Field

$$\tau = p \sin \theta$$

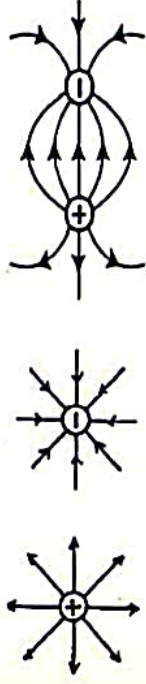
$$\vec{\tau} = \vec{p} \times \vec{E}$$

Electric Charges and Field



Properties of electric field lines

- 1.] Imaginary & Continuous curves.
- 2.] Starts from positive charge and end at negative charge.
- 3.] Do not form continuous closed loops.
- 4.] Always normal to the surface of conductor.
- 5.] Can never cross each other.



Gauss's Law

$$\Phi = \oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

by UMESH RAJORIA

Electric Field due to

- 1.] An infinite Line distribution of charge $E = \frac{\lambda}{2\pi\epsilon_0 r}$
- 2.] An Conducting or Hollow sphere $E_{\text{out}} = \frac{1}{4\pi\epsilon_0 R^2} \frac{q}{r^2}$ $E_{\text{in}} = 0$
- 3.] An infinite plane sheet of charge (non-conducting) $E = \frac{\sigma}{2\epsilon_0}$

Electric Dipole



Dipole Moment ($\vec{P}$)

$$\vec{P} = q \times 2\vec{a}$$

$$\vec{E}_{\text{axial}} = \frac{2K\vec{P}}{r^3}$$

$$\vec{E}_{\text{eqat}} = \frac{-K\vec{P}}{r^3}$$

$$E_{\text{axial}} = 2 E_{\text{equatorial}}$$

Electrostatic Potential Energy

$$U_P = W_{AP} = \int_A^P \vec{F}_{\text{ext}} \cdot d\vec{r} = - \int_A^P \vec{F}_E \cdot d\vec{r}$$

$$\Delta U = U_B - U_A = W_{AB} = \int_A^B \vec{F}_{\text{ext}} \cdot d\vec{r} = - \int_A^B \vec{F}_E \cdot d\vec{r}$$

Electrostatic Potential

$$V_P = \frac{W_{AP}}{q_0} \Rightarrow V_P = \frac{U_P}{q_0}$$

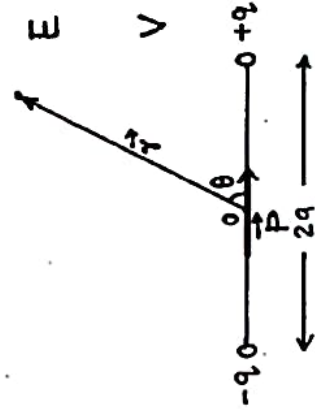
$$\Delta V = V_B - V_A = \frac{W_{AB}}{q_0} = \frac{U_B - U_A}{q_0}$$

SI Unit - Joule/Coulomb or Volt

Electric Potential at a point due to an Electric Dipole

$$E = \frac{kp\sqrt{3\cos^2\theta + 1}}{r^3}$$

$$V = \frac{kp\cos\theta}{r^2}$$



Electrostatic Potential due to a point charge

$$V = \frac{kq}{r}$$

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Electric Potential & Capacitance

Capacitors in series.

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Capacitors in parallel

$$C_p = C_1 + C_2 + C_3$$

Energy stored in a charged Capacitor

$$U = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2 = \frac{1}{2} QV$$

Potential Energy of a dipole in an external Field

$$\Delta U = U_2 - U_1 = W = -pE[\cos\theta_2 - \cos\theta_1]$$

$$U = -pE \cos\theta$$

Relation between Electric field and potential

$$E = -\frac{dV}{dr}$$

$$dV = -\vec{E} \cdot d\vec{r} \Rightarrow \Delta V = -\int_{r_1}^{r_2} \vec{E} \cdot d\vec{r}$$

Capacitor or Condenser

$$C = \frac{Q}{V}$$

Parallel plate Capacitor

$$C = \frac{\epsilon_0 A}{d}$$

$$C_{\text{medium}} = \frac{K\epsilon_0 A}{d}$$

When medium is partially present between the plates

$$C = \frac{\epsilon_0 A}{[d - t + \frac{t}{K}]}$$

Capacitance of a Spherical Conductor

$$C = 4\pi\epsilon_0 R$$

Electric Current $I = \frac{q}{t}$
 Instantaneous Current $I = \frac{dQ}{dt}$
 SI Unit - Ampere (A)
 Current is a scalar quantity.

Drift Velocity $V_d = \left(\frac{eE}{m}\right)\tau$

$$I = neAV_d$$

Mobility (μ) = $\frac{V_d}{E}$
 Current Density (J) = $\frac{I}{A}$

Resistors in Series

$$R_s = R_1 + R_2 + R_3$$

Resistors in parallel

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Ohm's Law

$$V = IR$$

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{m}{ne^2\tau}$$

Variation of Resistivity / Resistance with Temperature

For temperature variation from T_1 to T_2

$$R_2 = R_1 [1 + \alpha(T_2 - T_1)]$$

Current Electricity

Electrical Power

$$P = V \times I = I^2 R = \frac{V^2}{R}$$

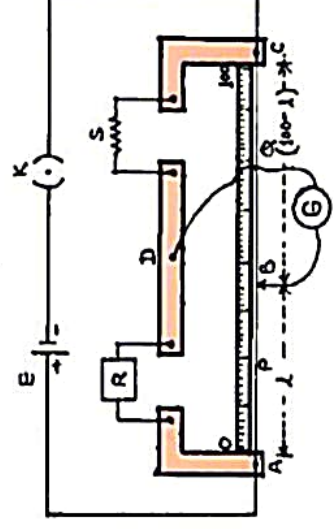
Electrical Energy

$$H = W = I^2 R t = \frac{V^2 t}{R}$$

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Meter Bridge

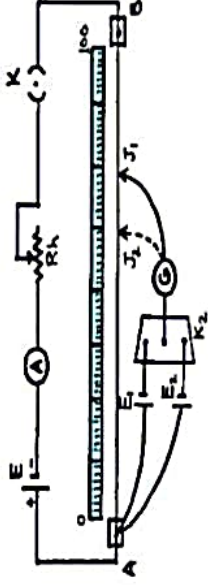
$$S = \left(\frac{100 - l}{l}\right) \times R$$



Potentiometer

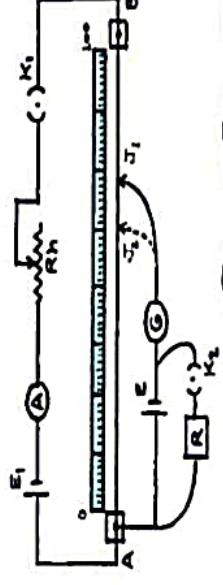
(A) Comparison of EMF's of two cells

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$



(B) Determination of internal resistance of the cell

$$r = \left(\frac{l_1}{l_2} - 1\right) R$$



Force on a moving charge in a magnetic field

$$F = qvB \sin \theta$$

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Magnetic Force on a current carrying conductor

$$F = ILB \sin \theta$$

$$\vec{F} = I \vec{l} \times \vec{B}$$

Motion in a Magnetic Field

(i) When charge particle moves at angle 90° to field

$$r = \frac{mv}{qB}$$

$$T = \frac{2\pi m}{qB}$$

(ii) When charge particle at any angle to the field

$$r = \frac{mv \sin \theta}{qB}$$

$$T = \frac{2\pi m}{qB}$$

$$\text{Pitch } P = \frac{2\pi mv \cos \theta}{qB}$$

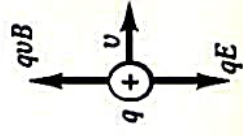
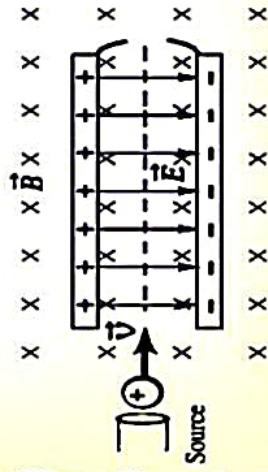
(iii) When the charged particle is moving at ($\theta = 0^\circ$ or 180°) to the magnetic field

$$F = 0$$

Path of the particle Linear (straight line)

Velocity Selector

$$v = \frac{E}{B}$$



Magnetic Effect of Current

BIOT SAVART LAW

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \hat{r}}{r^2}$$

(i) Magnetic Field on the axis of a Circular Current Loop

$$B = \frac{\mu_0 2\pi n I a^2}{4\pi (a^2 + x^2)^{3/2}}$$

at the centre of the circular coil

$$B_0 = \frac{\mu_0 n I}{2a}$$

(ii) Magnetic Field due to a Straight wire carrying current of infinite length

$$B = \frac{\mu_0 I}{2\pi a}$$

by Umesh Rajoria

Ampere's Circuital Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Magnetic Field due to a Solenoid

$$B = \mu_0 n I \quad n = \frac{N}{L}$$

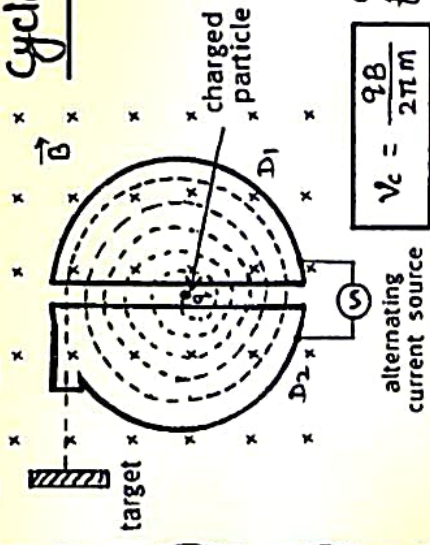
Magnetic Field due to TOROID

$$B = \mu_0 n I \quad n = \frac{N}{2\pi r}$$

Force between two parallel current carrying conductors

$$F = \frac{\mu_0 \times I_1 I_2}{2\pi y}$$

Cyclotron

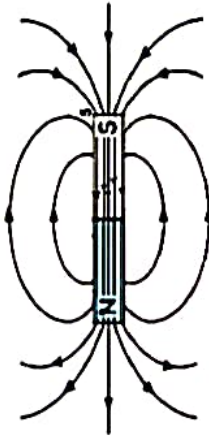


$$V_c = \frac{qB}{2\pi m} \quad \text{cyclotron frequency}$$

$$K.E. = \frac{1}{2} m v^2 = \frac{q^2 B^2 R^2}{2m}$$

Properties of Magnetic Field Lines

- Outside the magnet the field lines are from north pole to South pole



- field lines form closed continuous loops
- field lines cannot intersect each other
- magnitude of magnetic field is proportional to the density of magnetic field lines

Curie Law in Magnetism

$$\chi_m = \frac{C}{T}$$

Gauss's Law

$$\phi_B = \oint_S \vec{B} \cdot d\vec{S} = 0$$

Potential Energy of a Magnetic Dipole in a Magnetic Field

$$U = -MB \cos \theta$$

$$U = -\vec{M} \cdot \vec{B}$$

$$\Delta U = W = -MB (\cos \theta_2 - \cos \theta_1)$$

Magnetic Dipole Moment



$$\vec{M} = m(2\vec{l})$$

- its direction is from south to north pole

Magnetism and Matter

Magnetic Field due to Bar Magnet

On axial line

$$\vec{B} = \frac{\mu_0 2M}{4\pi r^3}$$

On equatorial line

$$\vec{B} = -\frac{\mu_0 M}{4\pi r^3}$$

Torque on a Bar Magnet in a Magnetic field

$$\tau = MB \sin \theta$$

$$\vec{\tau} = \vec{M} \times \vec{B}$$

(i) Relative magnetic permeability

$$\mu_r = \frac{\mu}{\mu_0}$$

$$\text{or } \mu_r = \frac{B}{B_0}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Wb A}^{-1} \text{ m}^{-1}$$

(ii) Magnetising Force or Magnetic Intensity

$$H = \frac{B_0}{\mu_0} \quad \text{or} \quad H = \frac{B}{\mu}$$

(iii) Magnetisation Intensity (I)

$$I = \frac{\text{Magnetic moment}}{\text{volume}} = \frac{M}{V}$$

$$I = \frac{m}{A} \quad I = \frac{B_m}{\mu_0}$$

(iv) Magnetic Susceptibility

$$\chi_m = \frac{I}{H}$$

$$\mu_r = 1 + \chi_m$$

χ_m , μ_r and μ of magnetic substances

Substance	χ_m	μ_r	μ
1. Diamagnetic	$-1 \leq \chi_m < 0$	$0 \leq \mu_r < 1$	$\mu < \mu_0$
2. paramagnetic	$0 < \chi_m < 1$	$1 < \mu_r < (1+\delta)$	$\mu > \mu_0$
3. Ferromagnetic	$\chi_m \gg 1$	$\mu_r \gg 1$	$\mu \gg \mu_0$

* Here δ is a small positive number.

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Hysteresis Loss $Q = V \Delta t$

n = frequency of cycles

t = time

V = Volume of the material

A = Area of hysteresis loop

Faraday's Law of EMI

First Law -

Whenever the magnetic flux changes an e.m.f. is induced in the circuit

Second Law -

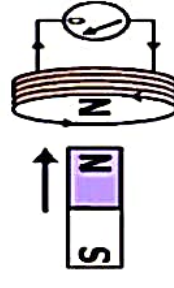
$$e = - \frac{d\phi}{dt}$$

$$I = - \frac{N}{R} \frac{d\phi}{dt}$$

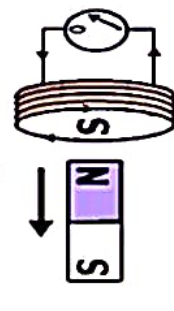
$$\Delta Q = - \frac{N}{R} \Delta \phi$$

Lenz's Law

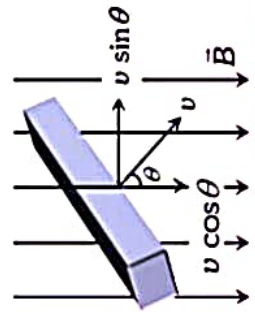
movement against repulsion



movement against attraction



Motional Electromotive Force



$$e = B l (v \sin \theta)$$

Self Inductance

$$L = \frac{\phi}{I}$$

$$e = - L \times \frac{dI}{dt}$$

various formulae for L

Circular coil

$$L = \frac{\mu_0 \pi N^2 r}{2}$$

Toroid

$$L = \frac{\mu_0 N^2 A}{l}$$

$$L = \frac{\mu_0 N^2 r}{2}$$

ELECTROMAGNETIC INDUCTION

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Mutual Inductance

$$\phi_2 = M I_1$$

$$e_2 = - M \frac{dI_1}{dt}$$

various formulae for M

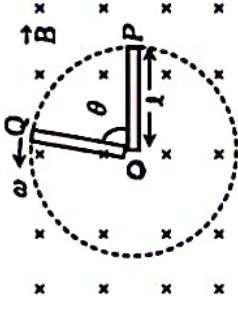
Two concentric coils

$$M = \frac{\pi \mu_0 N_1 N_2 r^2}{2R}$$

Two Solenoids

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

Rotating metallic rod



Induced e.m.f.

$$e = \frac{1}{2} B l^2 \omega$$

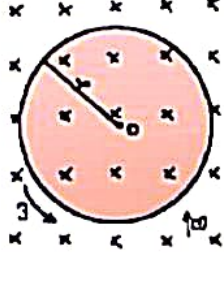
$$e = \pi v B l^2$$

Cycle wheel



$$e_{net} = e = \frac{1}{2} B l^2 \omega$$

Rotation of a metal disc



$$e = \frac{1}{2} B \omega r^2$$

Coefficient of Coupling

$$K = \frac{M}{\sqrt{L_1 L_2}}$$

$$0 \leq K \leq 1$$

Magnetic Energy of Inductor

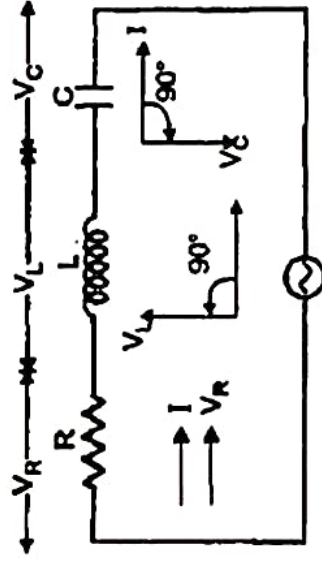
$$U = W = \frac{1}{2} L I^2$$

Advantages of A.C.

- A.C. is cheaper than D.C.
- A.C. can be easily converted into D.C. (by rectifier)
- It can be controlled easily (by choke coil)
- It can be transmitted over long distance at negligible power loss.
- It can be stepped up or stepped down with the help of transformer.

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Series (RLC circuit)



Impedance

$$Z = \frac{E_0}{I_0}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Phase diff.

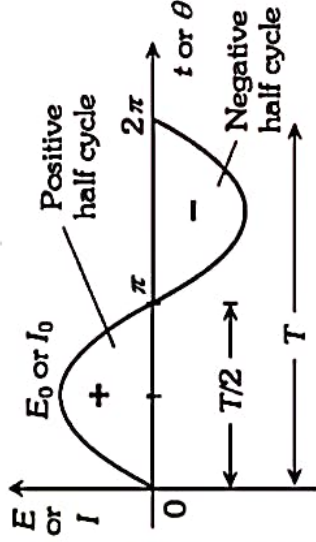
$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

Alternating Current

$$I = I_0 \sin \omega t$$

$$E = E_0 \sin \omega t$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$



Mean or Average value of A.C.

$$I_{av} = \frac{2I_0}{\pi} = 0.637 I_0$$

Root Mean Square Value of A.C.

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$$

Q - Factor

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Voltage across L or C

$$Q = \frac{\text{Voltage across L or C}}{\text{Applied Voltage (= voltage across R)}}$$

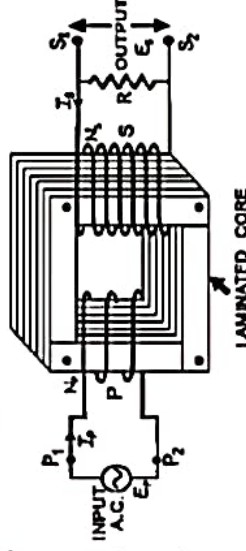
Average Power in RLC Circuit

$$P_{av} = E_{rms} I_{rms} \cos \phi$$

$$\text{Power factor} = \cos \phi = \frac{R}{Z}$$

Transformer

Principle - Mutual induction



$$\frac{E_s}{E_p} = \frac{N_s}{N_p}$$

For Ideal Transformer

$$\frac{I_p}{I_s} = \frac{E_s}{E_p}$$

Efficiency of transformer

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} \times 100 \%$$

Wattless Current
 $I_{rms} \sin \phi$

Wattful Current
 $I_{rms} \cos \phi$

Maxwell's Equations

(1)

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

(2)

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

(3)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{S}$$

(4)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot d\vec{S}$$

Displacement current $I_D = \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot d\vec{S}$

Lorentz Formula

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Production of Electromagnetic Waves

The EM waves are produced by the accelerated or oscillating charge.

$$c = \frac{E_0}{B_0}$$

$$B_y = B_0 \sin(\omega t + kx)$$

$$E_z = E_0 \sin(\omega t + kx)$$

$$\omega = \frac{2\pi}{T} \quad k = \frac{2\pi}{\lambda}$$

ELECTROMAGNETIC WAVES

Poynting Vector ($\vec{S}$)

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

Intensity $I = U_{av} \times C$

$$I = \frac{1}{2} \epsilon_0 E_0^2 C = \frac{1}{2} \frac{B_0^2}{\mu_0} C$$

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Average electric energy density

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

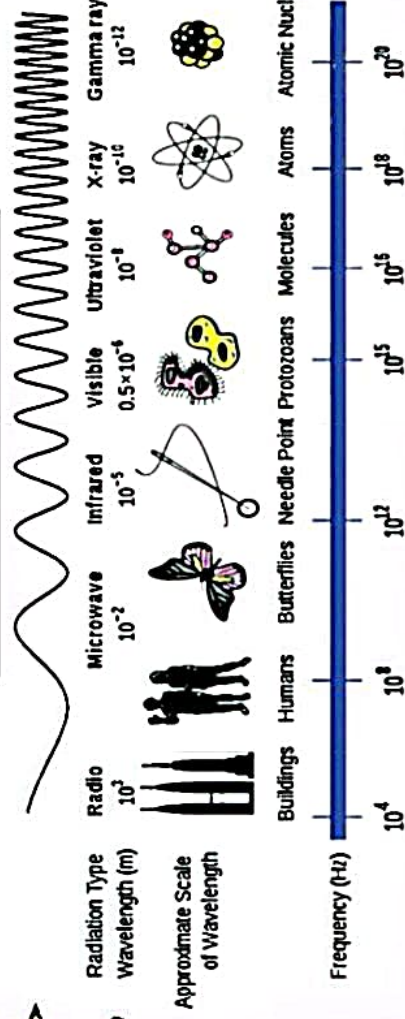
Average magnetic energy density

$$U_B = \frac{1}{2} \frac{B^2}{\mu_0}$$

Total energy per unit volume

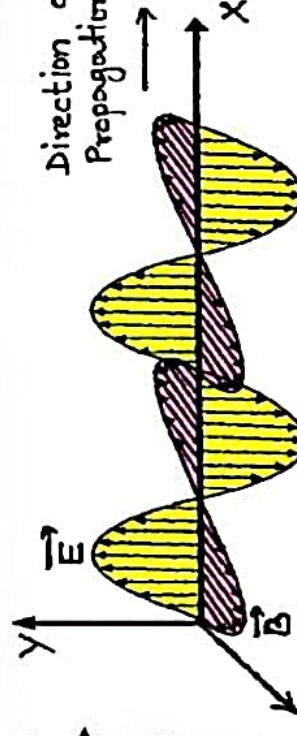
$$U_{av} = U_E + U_B$$

Electromagnetic Spectrum



Red Monster In Village is Ultimate example of Ghost

Direction of Propagation



$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

in vacuum

$$\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am}$$

$$v = \frac{1}{\sqrt{\mu \epsilon}}$$

in medium

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

Mirror Formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Linear Magnification $m = \frac{I}{O} = -\frac{v}{u}$

Object	Concave Mirror			Orientation
	Position	Image Size	Nature	
1. at ∞	at Focus	Point	Real	Inverted
2. between ∞ and C	between F and C	diminished	Real	Inverted
3. at C	at C	Same	Real	Inverted
4. between C and F	between C and ∞	Magnified	Real	Inverted
5. at F	at ∞	Indefinitely Large	Real	Inverted
6. between on same F and P side of Mirror		Large	Virtual	Erect

Object	Convex Mirror			Orientation
	Position	Image Size	Nature	
1. at ∞	at Focus	Point	Virtual	Erect
2. between ∞ and C	between Point F and C	diminished	Virtual	Erect

Lens Maker's Formula

$$\frac{1}{f} = (\mu_2 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Lens Formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Combination of Lenses in Contact

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Power of a Lens

$$P = \frac{1}{f}$$

RAY OPTICS

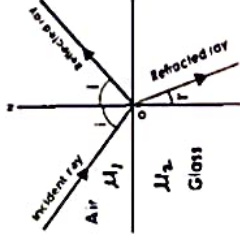
Refractive Index $\mu = \frac{c}{v}$

Relative Refractive Index

$$\mu_{21} = \frac{\mu_2}{\mu_1} = \frac{c/v_2}{c/v_1} = \frac{v_1}{v_2}$$

Snell's Law

$$\mu_1 \sin i = \mu_2 \sin r$$



Essential Conditions for total Internal Reflection

- (i) Light should travel from a denser medium to a rarer medium.
- (ii) Angle of incidence in denser medium should be greater than the critical angle.

$$\mu_b = \frac{1}{\sin c}$$

a = rarer

b = denser

c = critical angle

Spherical Refracting Surfaces

(i) Refraction from Rarer to Denser Medium

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

(ii) Refraction From Denser to Rarer Medium

$$\frac{\mu_2}{-u} + \frac{\mu_1}{v} = \frac{\mu_1 - \mu_2}{R}$$

Prism

Angle of deviation (δ)

$$\delta = (i_1 + i_2) - A$$

$$r_1 + r_2 = A$$

Prism formula

$$\mu = \frac{\sin \left(\frac{A + \delta}{2} \right)}{\sin \frac{A}{2}}$$

Compound Microscope

Magnifying power

$$m = -\frac{v_o}{u_o} \left(1 + \frac{d}{f_e} \right)$$

Length of microscope tube

$$L = v_o + u_e$$

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Resolving Power of telescope

$$R.P. \text{ of Telescope} = \frac{1}{d\theta} = \frac{D}{1.22\lambda}$$

D = diameter of the objective of the telescope
 λ = wavelength of light used.

Astronomical Telescope

(i) Normal Adjustment

Magnifying Power

$$m = \frac{f_o}{-f_e}$$

Length of the telescope

$$L = f_o + f_e$$

(ii) When the final image is formed at the least distance of distinct vision (d)

Magnifying Power

$$m = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{d} \right)$$

Length of the telescope

$$L = f_o + u_e$$

Resolving Power of Microscope

$$R.P. \text{ of microscope} = \frac{1}{d} = \frac{2\mu \sin \theta}{1.22\lambda}$$

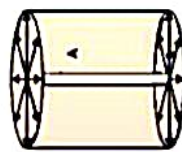
d = minimum distance between two objects

Wavefront

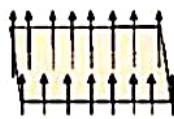
(i) Spherical wavefront



(ii) Cylindrical wavefront



(iii) Plane wavefront



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Wave Optics

Diffraction of Light at a single slit

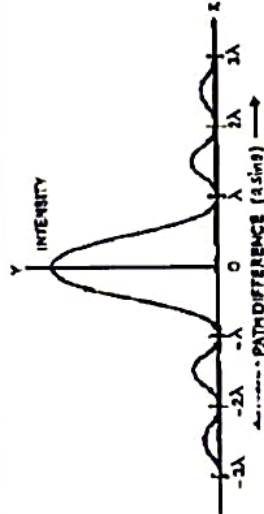
For n^{th} secondary minimum

$$a \sin \theta = n\lambda$$

for n^{th} secondary maximum

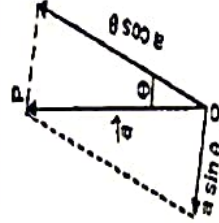
$$a \sin \theta = (2n+1)\frac{\lambda}{2}$$

$$\text{Width of central Maximum} = \frac{2D\lambda}{a}$$



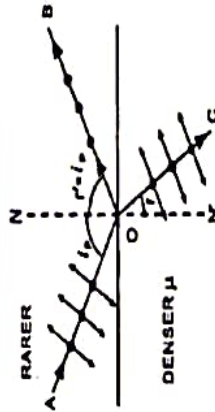
Law of Malus

$$I = I_0 \cos^2 \theta$$



Brewster's Law

$$\mu = \tan i_p$$



$$\text{path difference} = \frac{d \cdot x}{D}$$

For Constructive Interference

Position of Bright Fringe

$$x_n = \frac{n\lambda D}{d}$$

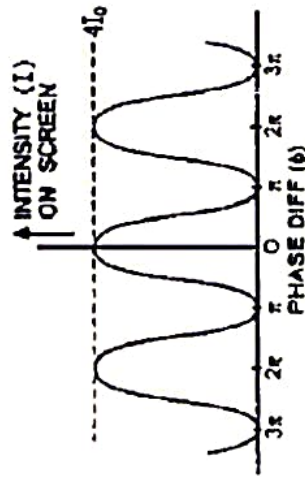
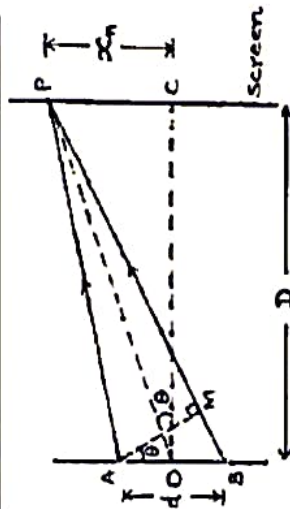
$$\text{Fringe width } \beta = \frac{\lambda D}{d}$$

For destructive Interference

$$x_n = \frac{(2n-1)\lambda D}{2d}$$

$$\text{Fringe width } \beta = \frac{\lambda D}{d}$$

Fringe width in Interference



Resultant Amplitude and intensity in Interference

$$y_1 = a_1 \sin \omega t$$

$$y_2 = a_2 \sin(\omega t + \phi)$$

Resultant amplitude

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

for bright fringe

$$\phi = 2n\pi$$

$$\Delta = n\lambda$$

for dark fringe

$$\phi = (2n-1)\pi$$

$$\Delta = (2n-1)\frac{\lambda}{2}$$